The Last Bench

Time limit per test: 1 second
Memory limit per test: 256 mb

Dr. Kashif Munir is a senior teacher at FAST Islamabad. Over the years he has taught more batches than he can count, and in that time he has developed a sixth sense for one particular species of student: the **Backbencher**.

Backbenchers are easy to spot. They arrive on time, nod politely, and then walk straight past every empty seat in the room until they reach the corner farthest from the teacher's desk. There, protected by three rows of classmates' heads, they open mobile games, play them with full concentration, and occasionally look up to say "Yes sir, understood."

Dr. Kashif decided he had seen enough. One Monday morning, he walked into class and announced a new rule, which he proudly named the **Zig-Zag Protocol**:

> **Dr. Kashif:** *"From today, nobody chooses their own seat. You will sit in the order you enter this room. No exceptions, no swapping, and no 'sir, my friend is saving this seat for me.'"*

The classroom has rows of exactly w seats each. Row 1 is the front row, and the teacher's desk sits at its **left end**, right next to the seat in row 1, column 1. Columns are numbered 1 to w from left to right, so column 1 is always the side closest to the desk.

The students who enter the room are seated as follows:

1. The 1st student to enter sits in row 1, column 1, directly in front of the desk. Students call this seat "the Hot Seat." Its occupant can read Dr. Kashif's handwriting before he even writes it.
2. The next students fill row 1 **from left to right**, until row 1 is full.
3. Then row 2 is filled **from right to left**, starting at column w.
4. Then row 3 is filled from left to right, row 4 from right to left, and so on. The direction flips with every row, like a snake winding its way to the back of the room.

Seats are always filled in exactly this order, with no gaps. If n students attend, the first n seats of the snake are occupied and the rest stay empty. The last occupied row may therefore be only partly filled. The room always has enough rows for everyone.

For example, with $w = 4$ and 10 students, the seating looks like this (numbers show the order in which each student entered):

```
 DESK
  |
Row 1:   1   2   3   4
Row 2:   8   7   6   5
Row 3:   9  10   .   .
```

Dr. Kashif was satisfied. *"Let them try to sit in the corner now,"* he said, sipping his tea.

Shahid is a very clever student who would do anything to avoid Dr. Kashif's line of sight. Unfortunately, he cannot simply stop coming to class. The university's **80% attendance policy** hangs over him like a sword, and his attendance currently stands at 80.4%, which Shahid describes as "a comfortable lead." One more absence and he gets debarred.

So Shahid must attend. But he noticed something about the Zig-Zag Protocol: **the seat depends only on the order of entry**. If he waits outside the door and counts how many students have entered, he can walk in at exactly the right moment and become the k**-th** student to enter, for any k he likes from 1 to n. The other students enter in the remaining positions, and their order doesn't matter to him.

Shahid measures how far a seat is from the teacher with what he calls the **Glare Distance**. The seat in row i, column j has

$$\text{Glare Distance} = (i-1) + (j-1).$$

In other words, it is the number of rows behind the front row plus the number of columns away from the desk side. The Hot Seat has a Glare Distance of 0. The larger the Glare Distance, the safer Shahid's games are.

Shahid wants to choose k so that his seat has the **maximum possible Glare Distance**. If several values of k give the same maximum distance, he picks the **largest** such k, because arriving later means sleeping longer.

Shahid's first instinct was to always come in last. After all, the last student sits in the last row, right? Before he bets his attendance on that, help him find the correct answer.

Input:

The first line contains a single integer T, the number of lecture days Shahid needs to plan for.

Each of the next T lines contains two integers $\boldsymbol{n}$ and $\boldsymbol{w}$: the number of students attending that lecture (including Shahid) and the number of seats in each row of that day's classroom.

Output:

For each lecture day, print a single integer k on its own line: the position in which Shahid should enter the classroom.

Constraints:

- $1 \leq T \leq 10^5$
- $1 \leq n \leq 10^{18}$
- $1 \leq w \leq 10^{18}$

Note that n and w can be very large, because FAST has a lot of students, and Dr. Kashif has a lot of patience.

Example:

Example 1

Input

```
4
6 3
10 4
11 4
12 4
```

Output

```
4
5
11
12
```

Explanation:

Day 1 ($n = 6$, $w = 3$):

```
 DESK
  |
Row 1:   1   2   3
Row 2:   6   5   4
```

Row 2 is filled from right to left, so the 4th student gets the back-right corner: row 2, column 3, with a Glare Distance of $1 + 2 = 3$. The last student (6th) ends up in row 2, column 1, with a Glare Distance of only 1, right beside the desk side of the room. Shahid's "come in last" plan would have been a disaster.

Day 2 ($n = 10$, $w = 4$):

```
 DESK
  |
Row 1:   1   2   3   4
Row 2:   8   7   6   5
Row 3:   9  10   .   .
```

The 10th student sits in row 3, column 2, with a Glare Distance of $2 + 1 = 3$. The 5th student sits in row 2, column 4, with a Glare Distance of $1 + 3 = 4$. The partly filled last row is too short to beat the corner of the row in front of it, so Shahid should enter 5th.

Day 3 ($n = 11$, $w = 4$):

```
 DESK
  |
Row 1:   1   2   3   4
Row 2:   8   7   6   5
Row 3:   9  10  11   .
```

The 5th student (row 2, column 4) and the 11th student (row 3, column 3) both have a Glare Distance of 4. It is a tie, so Shahid picks the later one and enjoys some extra sleep: $k = 11$.

Day 4 ($n = 12$, $w = 4$):

```
 DESK
  |
Row 1:   1   2   3   4
Row 2:   8   7   6   5
Row 3:   9  10  11  12
```

This time the last row is full and filled from left to right, so the 12th student gets the back-right corner with a Glare Distance of $2 + 3 = 5$. Coming in last finally works.